NIHARIKA SINGH

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Professional Summary

AI/ML Engineer skilled in Machine Learning, NLP, Generative AI, and Python development with hands-on experience building RAG-based applications, sentiment analysis systems, and ML-powered solutions. Experienced in deploying scalable applications using Streamlit Cloud and integrating LLM workflows with vector databases and APIs. Strong analytical, problem-solving, and software development skills with an interest in AI, cloud technologies, and enterprise solutions.

Core Skills

- **Languages:** Python, SQL, R Programming
- **ML / AI:** TensorFlow, PyTorch, LangChain, LangGraph, LangSmith, FAISS, Google Gemini API, Scikit-learn
- **Cloud & Deployment:** AWS Fundamentals, Microsoft Azure Fundamentals, Streamlit Cloud, REST APIs, API Integration
- **LLM & NLP:** Embeddings, Retrieval-Augmented Generation (RAG), Tokenisation, Vectorisation
- **Libraries & Tools:** Pandas, NumPy, Matplotlib, Scikit-learn, Git, GitHub
- **Development:** HTML, CSS, JavaScript, React.js, Flask
- **Databases:** FAISS, Basic SQL Databases

Projects

RAG-Based Document Assistant

- Engineered a Retrieval-Augmented Generation (RAG) assistant capable of answering user queries across multiple document formats, including PDF, DOCX, TXT, and CSV.
- Integrated Gemini embeddings with FAISS-based semantic search to improve retrieval relevance and response efficiency.
- Optimised large language model workflows through strategic chunking and tokenisation techniques.
- Deployed the application on Streamlit Cloud to support scalable and real-time user interaction.

Sentiment Analysis

- Designed a sentiment analysis classifier for social media data using NLP preprocessing and machine learning techniques.
- Optimized text preprocessing and segmentation workflows to improve pipeline efficiency.
- Automated data cleaning and preprocessing processes for faster sentiment analysis execution.
- Visualized sentiment trends and distributions to derive actionable insights.

Image Enhancer

- Developed an image enhancement platform using ML-driven preprocessing methods to improve image clarity and quality.
- Integrated React.js frontend with Flask backend to build a scalable and production-ready workflow.
- Tested the application on multiple image samples to ensure consistent processing across varying resolutions.
- Standardized preprocessing stages to improve output consistency and reduce model variance.

Education

Shri Shankaracharya Technical Campus, Bhilai – B.Tech Computer Science Engineering

GPA: 8.7 | Aug 2021 – Aug 2025

Awards

- **MLH Hackathon – Domain Name Track Challenge Winner (Jan 2024)**

Developed and presented an AI-based solution that won the MLH Domain Name Track Challenge.